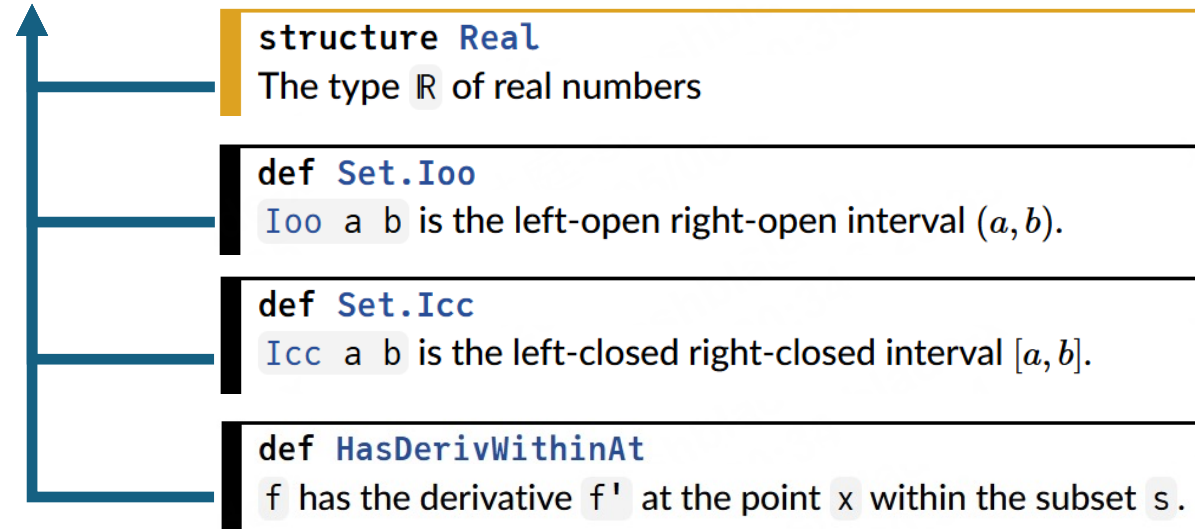


```

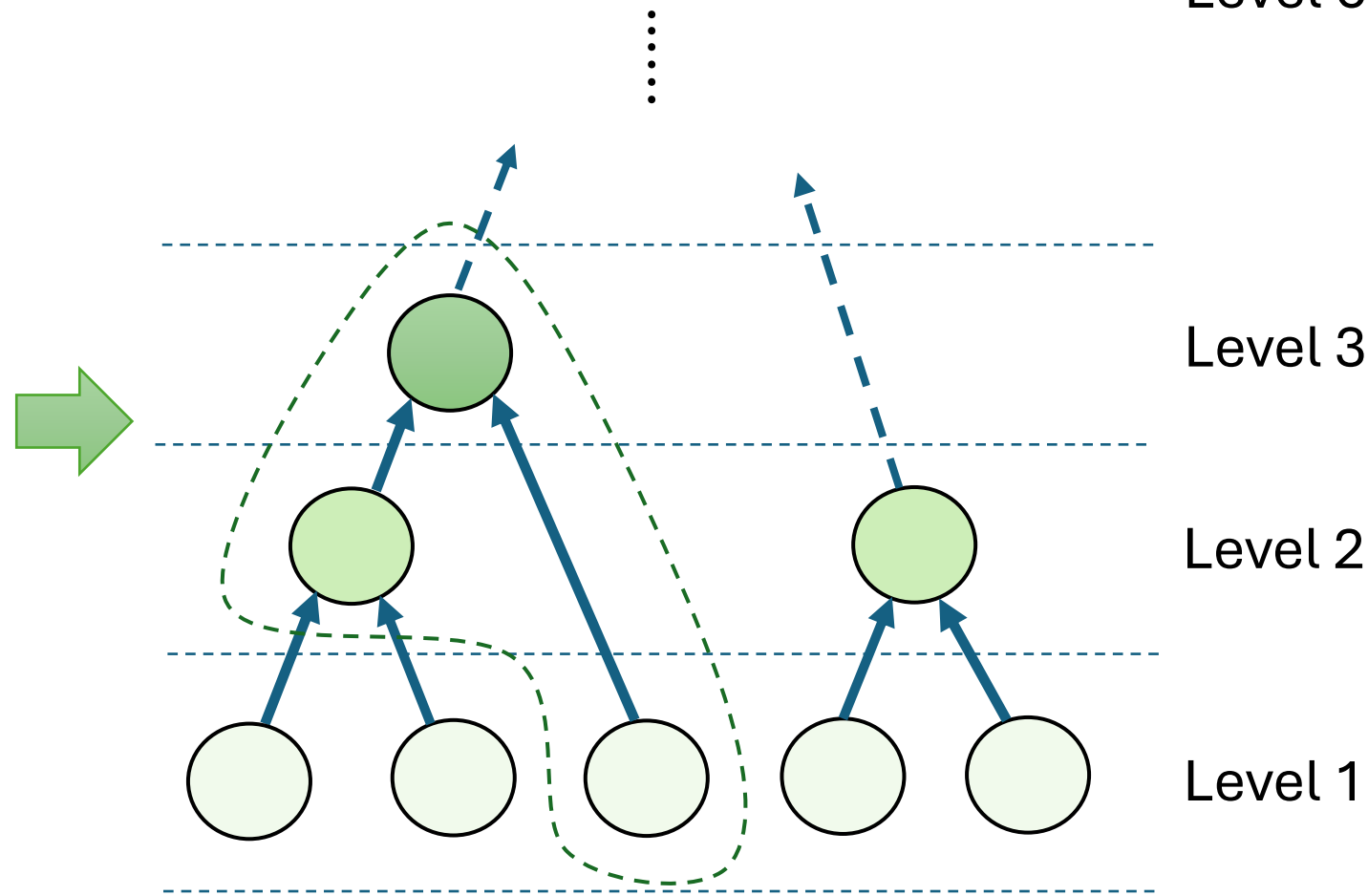
theorem exists_hasDerivWithinAt_eq_of_gt_of_lt
  {a b : ℝ} {f f' : ℝ → ℝ} (hab : a ≤ b)
  (hf : ∀ x ∈ Set.Icc a b, HasDerivWithinAt f (f' x) (Set.Icc a b) x)
  (hmb : m < f' b) (hma : f' a < m)
  m ∈ f' '' Set.Ioo a b

```

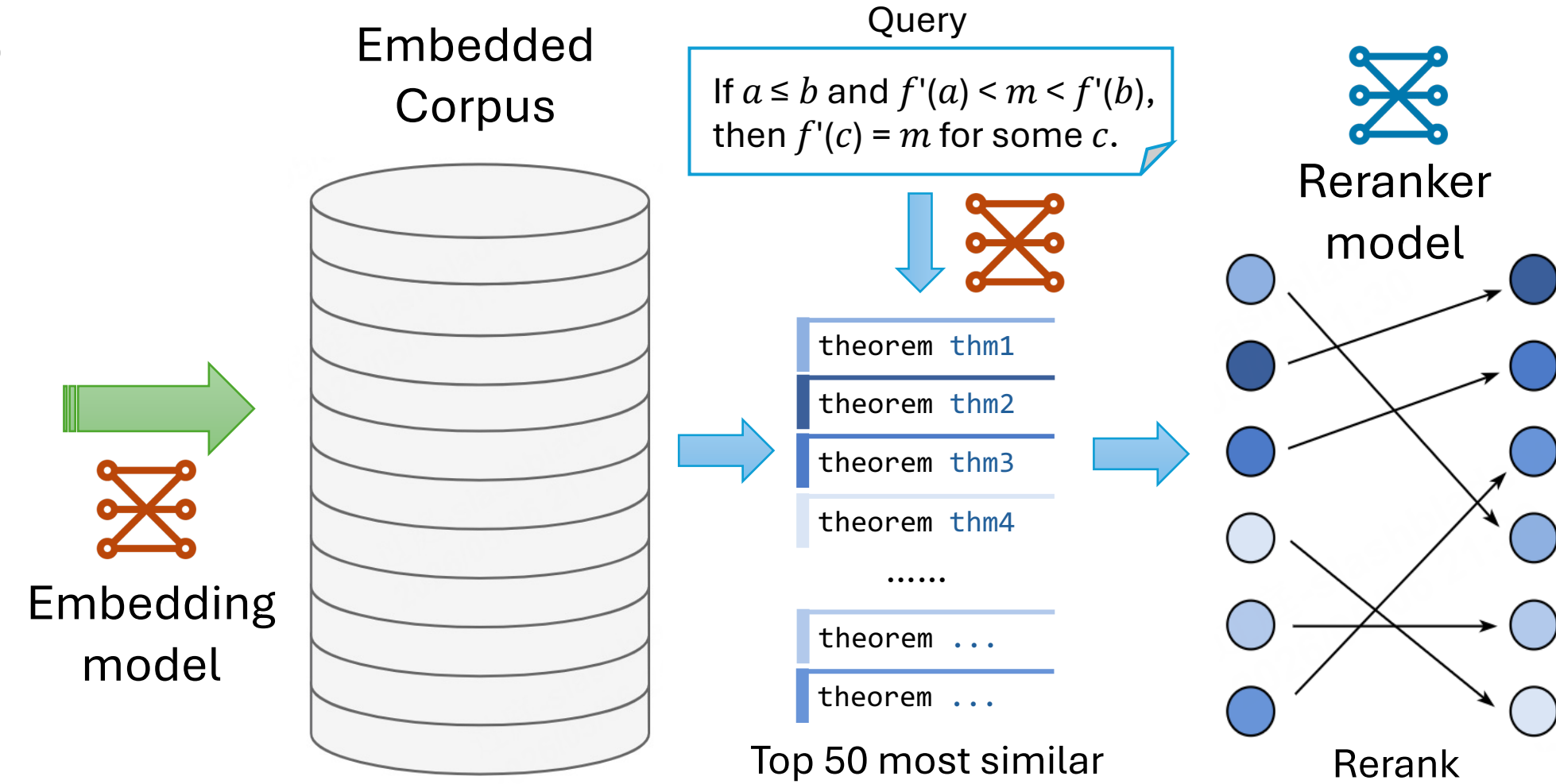
Darboux's theorem: if $a \leq b$ and $f' a < m < f' b$, then $f' c = m$ for some $c \in (a, b)$



(a) Local dependency extraction



(b) Bottom-up hierarchy informalization



(c) Embedding-reranker retrieval pipeline